Study XYZ-001: Safety & Demographics Report

Integrated Clinical Trial Summary — Example

Statistical Programming, Biostatistics

14 October 2026

Introduction

This report summarises the **demographics** and **adverse event** profile for Study XYZ-001, a Phase II randomised controlled trial comparing **Drug A** versus **Placebo** in adult subjects with condition Z.

> **Note for programmers:** This document is a single `.qmd` source file. The tables, figures, and narrative are produced from the same R code every time the document is rendered — no copy-paste from SAS output.

Data

Create Fictional Datasets

In a real study environment these datasets would be read from SAS transport files (`.xpt`) or from a validated ADaM data store on the clinical network. Here we create small fictional datasets inline so the example is **fully self-contained**.

```
# A tibble: 2 × 2
  TRT01P      n
  <chr>   <int>
1 Drug A     30
2 Placebo    30
```

```
[1] 75
```

Demographics Summary

The table below mirrors what you would produce in SAS with `PROC TABULATE` or `PROC REPORT`. Here it is built with `dplyr` and rendered with `kable()`.

Table 1: Table 1. Subject Demographics and Baseline Characteristics (Safety Population)

Characteristic	Drug A (N=30)	Placebo (N=30)
Age (years), mean (SD)	52.8 (13.8)	48.8 (10.5)
Weight at baseline (kg), mean (SD)	74.3 (11.8)	77.8 (12.8)
Female, n (%)	19 (63.3%)	13 (43.3%)
White, n (%)	18 (60.0%)	23 (76.7%)
Black or African American, n (%)	6 (20.0%)	0 (0.0%)
Asian, n (%)	5 (16.7%)	5 (16.7%)

Adverse Events

AE Incidence by Treatment Arm

Table 2: Table 2. Subjects with Any Adverse Event by System Organ Class and Preferred Term

Adverse Event (Preferred Term)	Drug A (N=30)	Placebo (N=30)
Back pain	3 (10%)	2 (6.7%)
Dizziness	8 (26.7%)	2 (6.7%)
Fatigue	5 (16.7%)	3 (10%)
Headache	4 (13.3%)	4 (13.3%)
Insomnia	5 (16.7%)	4 (13.3%)
Nausea	5 (16.7%)	5 (16.7%)
Rash	7 (23.3%)	4 (13.3%)
Vomiting	6 (20%)	4 (13.3%)

AE Severity Profile

The figure below is produced inline by R — no external image file needed. In SAS you would export an ODS Graphics PNG and link it manually. Here the plot is part of the reproducible document.

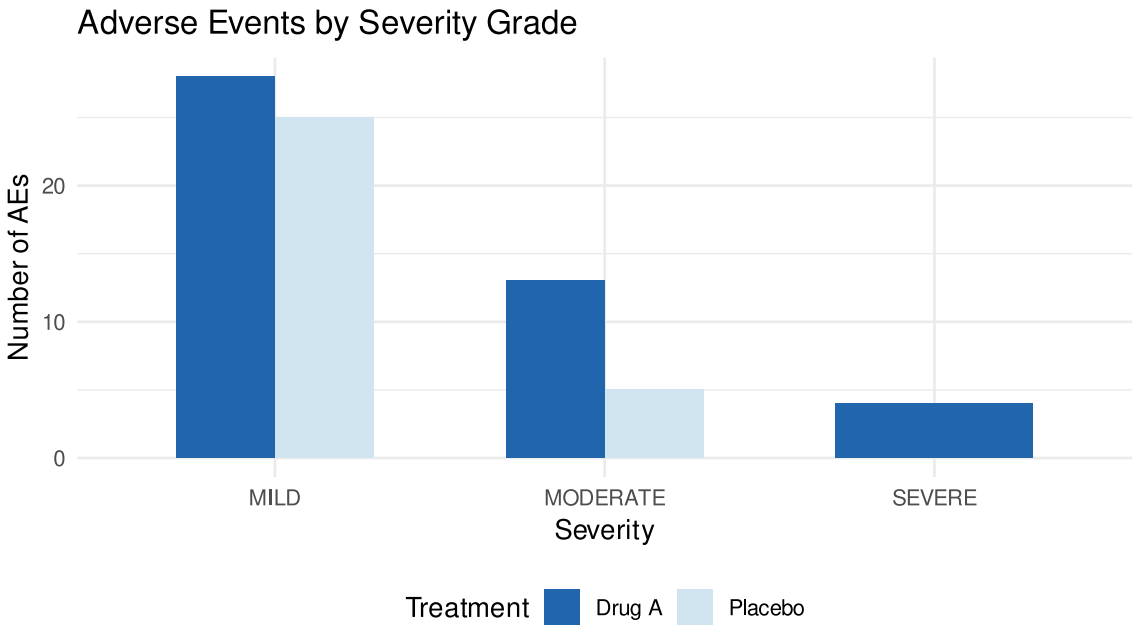


Figure 1: Figure 1. Adverse Event Counts by Severity and Treatment Arm

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Inline R Values in Narrative Text

One of Quarto's most powerful features for clinical programming: values computed in R can be embedded directly in prose.

> A total of **60** subjects were included in the Safety Population: **30** in the Drug A arm and **30** in the Placebo arm. Across both arms, **45** subjects (75%) experienced at least one adverse event.

If the data change upstream and you re-render, **every number updates automatically**. No more manual edits after an unblinding data cut.

Key Takeaways for SAS Programmers

SAS concept	Quarto / R equivalent
`ODS RTF` / `ODS PDF`	Change `format:` in YAML header
`%include 'setup.sas'`	`source("_setup.R")` or `library()`
`PROC REPORT` / `PROC TABULATE`	`dplyr` + `kable()` / `gt` / `flextable`
`ODS Graphics`	`ggplot2` — inline, no external file
Hard-coded counts in text	`60` — live inline values
Manual re-run after data cut	`quarto render` — full refresh in one command

Appendix: Session Info

Good practice: capture the R environment for reproducibility traceability — equivalent to logging SAS version and macro versions.

```
R version: R version 4.6.1 (2026-06-24)
```

```
Platform: x86_64-pc-linux-gnu
```

```
Key packages loaded:
```

```
  dplyr 1.2.1
  ggplot2 4.0.3
  knitr 1.52
  tidyr 1.3.2
```

```

Rendered: 2026-10-14 22:18:45 UTC
```